

Tutorial Quiz 7 Rubric

Q1]

$$\text{velocity, } \vec{v}(t) = \frac{d\vec{r}}{dt} = (-a \sin t) \hat{i} + (a \cos t) \hat{j} + b \hat{k}$$

$$\text{acceleration, } \vec{a}(t) = \frac{d\vec{v}}{dt} = (-a \cos t) \hat{i} + (-a \sin t) \hat{j}$$

$$\begin{aligned} |\vec{v}| &= \sqrt{a^2 \sin^2 t + a^2 \cos^2 t + b^2} \\ &= \sqrt{a^2 + b^2} \end{aligned}$$

$$\begin{aligned} |\vec{a}| &= \sqrt{a^2 \cos^2 t + a^2 \sin^2 t} \\ &= a \end{aligned}$$

$$\text{tangential component } a_T = \frac{d}{dt} |\vec{v}|$$

Since $|\vec{v}| = \sqrt{a^2 + b^2}$, it is a constant

$$\therefore a_T = 0$$

$$\begin{aligned} \text{normal component } a_N &= \sqrt{|\vec{a}|^2 - a_T^2} \\ &= \sqrt{a^2 - 0} \\ &= a \end{aligned}$$

$$\begin{aligned}
 \therefore a &= a_T T + a_N N \\
 &= 0 \cdot T + a_N N \\
 &= a_N N \quad \text{--- (1)}
 \end{aligned}$$

Q2]

$$v(t) = \frac{dr}{dt} = 2tj + 2k$$

$$\begin{aligned}
 |v(t)| &= \sqrt{(2t)^2 + 2^2} \\
 &= 2\sqrt{t^2 + 1}
 \end{aligned}$$

$$\begin{aligned}
 \text{On the curve, } x &= 0; \\
 y &= t^2 - 1 \\
 z &= 2t
 \end{aligned}$$

$$\text{density, } \delta = 15\sqrt{t^2 + 1}$$

$$\begin{aligned}
 M &= \int_{-1}^1 15\sqrt{t^2 + 1} \cdot 2\sqrt{t^2 + 1} \, dt \\
 &= 30 \int_{-1}^1 (t^2 + 1) \, dt \\
 &= 30 \left[\frac{t^3}{3} + t \right]_{-1}^1 \\
 &= 30 \left[\left(\frac{1}{3} + 1 \right) - \left(-\frac{1}{3} - 1 \right) \right] \\
 &= 30 \cdot \frac{8}{3} = 80 \quad \text{--- (1)}
 \end{aligned}$$

i) By symmetry $\bar{x} = 0$ (curve lies on yz plane)

ii) For $\bar{y}$, the integrand $y \cdot \delta \cdot |v(t)|$ is

$$\Rightarrow (t^2 - 1) \cdot 15 \sqrt{t^2 + 1} \cdot 2 \sqrt{t^2 + 1}$$

$$\Rightarrow 30 (t^2 - 1) (t^2 + 1)$$

$$\Rightarrow 30 (t^4 - 1)$$

$$\begin{aligned} \therefore M_{yz} &= 30 \int_{-1}^1 (t^4 - 1) dt \\ &= 30 \left[\frac{t^5}{5} - t \right]_{-1}^1 = -48 \end{aligned}$$

$$\bar{y} = \frac{M_{yz}}{M} = \frac{-48}{80} = -\frac{3}{5}$$

iii) For $\bar{z}$, the integrand $z \cdot \delta \cdot |v(t)|$

$$\Rightarrow 2t \cdot 30 (t^2 + 1) = 60t (t^2 + 1)$$

which is an odd y^n

$$M_{xz} = 0 \Rightarrow \bar{z} = 0$$

$$\therefore \text{center of mass} = \left(0, -\frac{3}{5}, 0 \right) \quad \text{--- (1)}$$